TMA4315 Exercise 1 Linear models for Gaussian data

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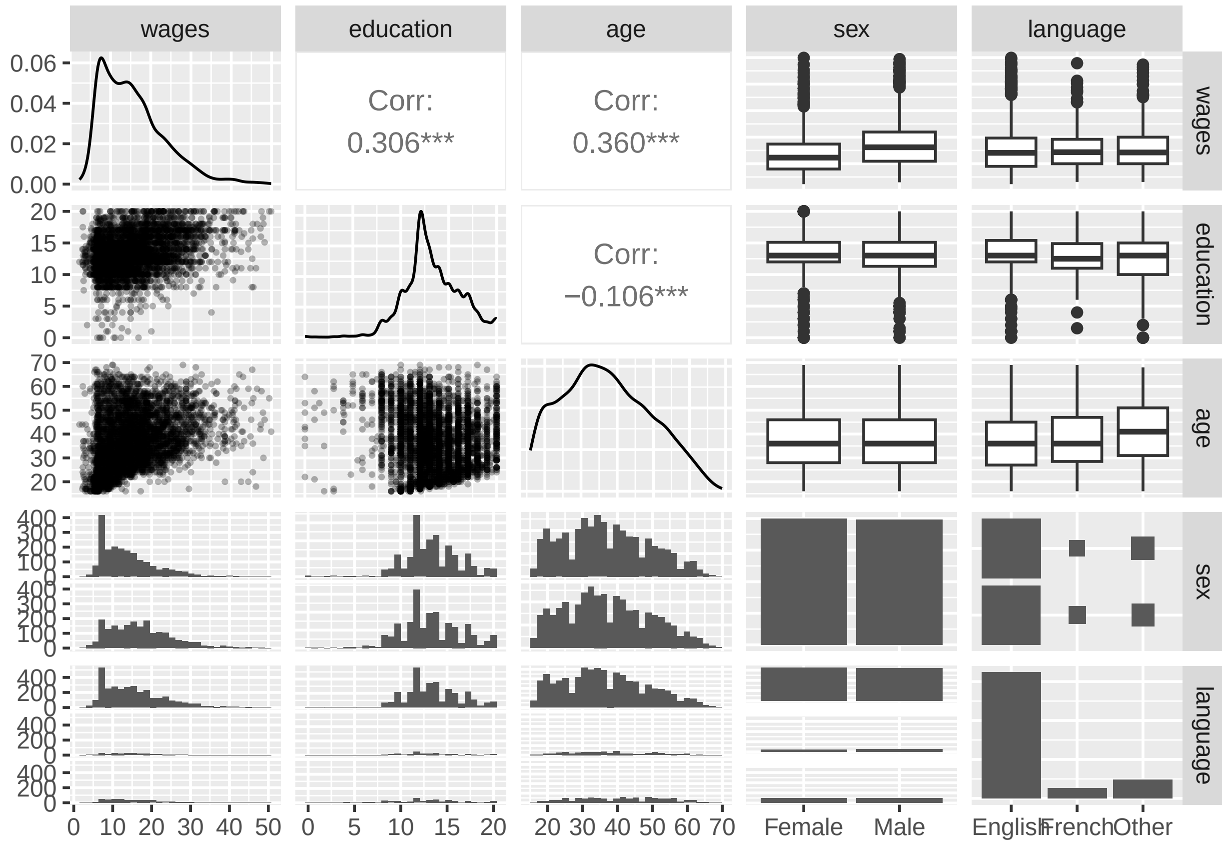
Contents

#Part 1: Explanatory analysis of the dataset First of all we import the dataset and prepare the data for the analysis

```
library(car)
data(SLID, package = "carData")
SLID <- SLID[complete.cases(SLID), ]
```

Afterwards we draw the draw a matrix of diagnostic plots of all the variables

```
ggpairs(SLID, lower = list(continuous = wrap("points", alpha = 0.3, size=0.5)))
```



Looking at the plot, there appear to be some noteworthy relationships between pairs of variables:

- `education` and `wages` have a moderate positive correlation of 0.306 that is, however, highly statistically significant. The same applies to `age` and `wages`.
- median wages are noticeably higher for male individuals compared to females, while they are effectively the same in relation to the language. This suggests that an individual's gender might be strongly related with their wage, while their language has a likely much weaker effect. In both cases, however, we can see that most wages are concentrated towards relatively low values, with the exception of a few, much higher outliers. This is further confirmed by observing the `wages` density plot, which is skewed to the right.
- `age` and `education` have a very weak, but highly statistically significant, negative correlation of -0.106. This means that these two variables are most likely uncorrelated.
- median education is effectively the same for male and female individuals, and very similar for individuals of different languages. This intuitively suggests that education is unaffected by these two variables. In both cases, the median education is relatively high, with the exception of a few outliers with a significantly lower value.
- a similar argument applies to the `age` variable, although it is worth noticing that the median age of individuals of `other` language appears to be higher.
- there does not seem to be anything noteworth about the relationship between age and language. It can be noticed, however, that the split between male and female individuals seems almost equal, while the `language` variables is dominated by `English`.

To carry out a multiple linear regression analysis of y on the covariates $x_1, \dots, x_p$ we need the following assumptions.

- **Linearity and additivity.** The expected response is a linear function of the regression parameters and the effects of the covariates add up:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i, \qquad i = 1, \ldots, n,$$

  or in matrix form $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$.

- **Homoscedasticity.** The errors have a constant variance, $\mathrm{Var}(\varepsilon_i) = \sigma^2$ for all i, which does not depend on the covariates.

- **Uncorrelated errors.** $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, so that $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$.

- **Full rank design.** The design matrix $\mathbf{X}$ has full column rank $p < n$, so that $\mathbf{X}^\top \mathbf{X}$ is invertible and $\hat{\boldsymbol{\beta}}$ is uniquely defined. The covariates are also assumed to be fixed (measured without error).

- **Normality.** For the *normal* linear regression model we additionally assume $\varepsilon_i \sim N(0, \sigma^2)$, independently. This is what makes the exact t- and F-tests valid.

Part 2: Linear regression with the mylm package

Statistical Formulas and Theory

The `mylm` package implements ordinary least squares (OLS) for a linear model $Y = X\beta + \varepsilon$. The functions in the package are built upon the following formulas:

- **Coefficient estimates:** $\hat{\beta} = (X^\top X)^{-1} X^\top Y$
- **Fitted values:** $\hat{Y} = X\hat{\beta}$
- **Residuals:** $e = Y - \hat{Y}$
- **Residual Sum of Squares (SSE):** $SSE = e^\top e$
- **Degrees of Freedom (df):** $df = n - p$ (where p is the number of parameters).
- **Total Sum of Squares (SST):** $SST = \sum(y_i - \bar{y})^2$
- **Covariance matrix:** $\widehat{\mathrm{Cov}}(\hat{\beta}) = \hat{\sigma}^2 (X^\top X)^{-1}$, with $\hat{\sigma}^2 = \frac{SSE}{n-p}$
- **Standard Errors (SE):** $\mathrm{SE}(\hat{\beta}_j) = \sqrt{\widehat{\mathrm{Cov}}(\hat{\beta})_{jj}}$
- **Asymptotic z-test:** $z_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}$ with p-value $2\Phi(-|z_j|)$.
- **Asymptotic** χ^2-test: $\chi^2 = \frac{SSE_{\text{reduced}} - SSE_{\text{full}}}{\hat{\sigma}^2_{\text{full}}}$
- **Coefficient of determination:** $R^2 = 1 - \frac{SSE}{SST}$

a) & b) Fitting the model and calculating standard errors

We fit a simple linear regression model to the data with `wages` as the response and `education` as the predictor.

```r
model1 <- mylm(wages ~ education, data = SLID)
model1b <- lm(wages ~ education, data = SLID)

print.mylm(model1)
```

```
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
## (Intercept)   education
##   4.9716912   0.7923091
```

```r
print(model1b)
```

```
##
## Call:
## lm(formula = wages ~ education, data = SLID)
##
## Coefficients:
## (Intercept)    education
##      4.9717       0.7923
```

The output confirms that our `mylm` function yields the exact same coefficient estimates as the standard `lm` function.

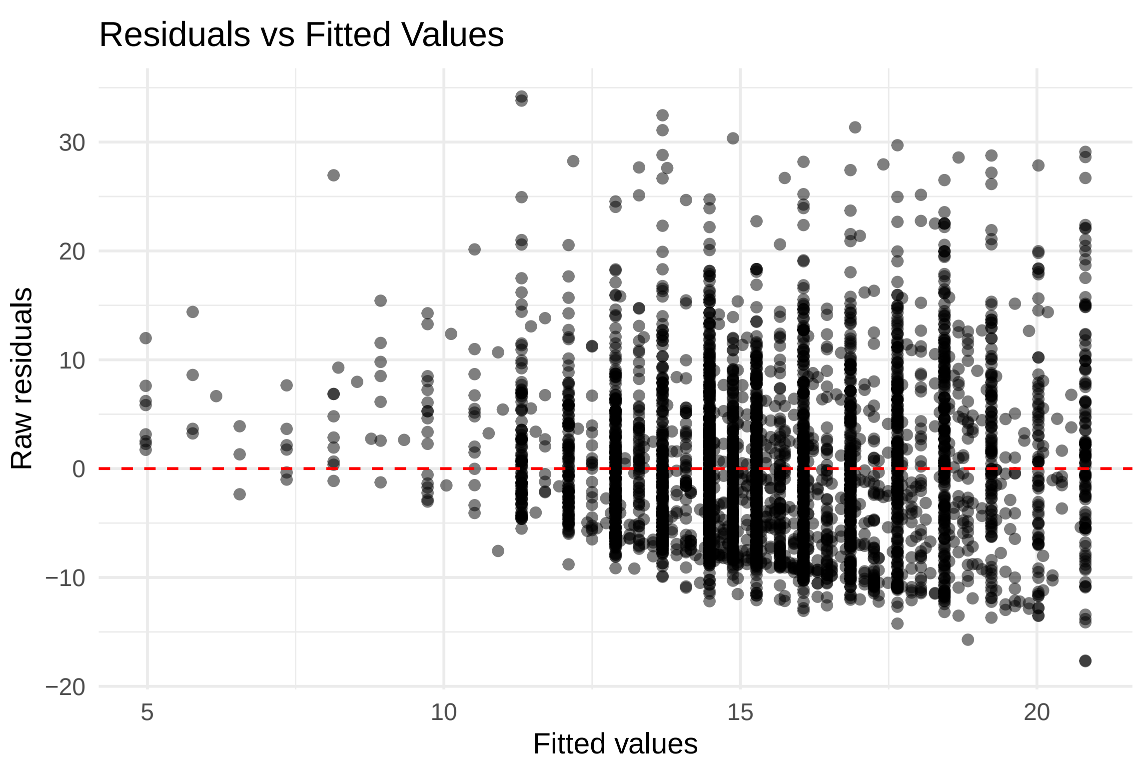
```r
summary.mylm(model1)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
##              Estimate Std. Error z value  Pr(>|z|)
## (Intercept)  4.97169    0.53429  9.3052 < 2.2e-16 ***
## education    0.79231    0.03906 20.2842 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.093586
```

Interpretation of parameters: * **Intercept ($\hat{\beta}_0 \approx 4.97$):** The expected hourly wage for a person with 0 years of education. This is extrapolated outside the data range and holds no practical meaning since wages cannot be negative. * **Education ($\hat{\beta}_1 \approx 0.79$):** For each additional year of education, the expected hourly wage increases by 0.79 units. * **Significance (z-test):** The z-value for education is 20.28, yielding a p-value strictly close to 0. We confidently reject the null hypothesis ($H_0 : \beta_1 = 0$); education is a highly significant predictor.

c) Residuals plot

```r
plot.mylm(model1)
```



Comment on the plot: The residuals versus fitted values plot shows a funnel shape, meaning the variance of the residuals increases as the fitted values increase. This violates the assumption of homoscedasticity.

d) ANOVA and Chi-squared test

```r
results_anova <- anova.mylm(model1)
```

```
## Analysis of Variance Table
## Response: wages
##           Df  Sum sq X2 value Pr(>X2)
## education 1   23096.20 411.45   1.7747e-91
```

Relationship between tests: In simple linear regression, the χ^2 statistic with 1 degree of freedom is equal to the square of the z-statistic for the slope coefficient. From our summary, $z \approx 20.284$, and $z^2 \approx 411.45$, matching the χ^2 value in the ANOVA table. The critical value for a 5% significance level is $z_{0.025} \approx 1.96$ and $\chi^2_{1,0.05} \approx 3.84$.

e) Coefficient of determination (R^2)

The coefficient of determination R^2 is **0.0936**.

Interpretation: Approximately **9.36%** of the total variability in hourly wages is explained by the variance in years of education. The remaining **90.64%** is unexplained by this model.

Part 3: Multiple linear regression

a) Fitting the multiple regression model

Because our `mylm` and `print.mylm` functions were strictly implemented using generalized matrix operations (specifically $\hat{\beta} = (X^\top X)^{-1} X^\top Y$), they automatically adapt to any number of predictors. No changes to the underlying R functions are necessary to handle multiple linear regression.

```r
model2 <- mylm(wages ~ education + age, data = SLID)
print.mylm(model2)
```

```
##
## Call:
## mylm(formula = wages ~ education + age, data = SLID)
##
## Coefficients:
## (Intercept)   education         age
##  -6.0216529   0.9014644   0.2570898
```

b) Estimates, standard errors, and z-test

Similarly, our `summary.mylm` function uses the generalized covariance matrix formula $\hat{\sigma}^2(X^\top X)^{-1}$. Extracting the diagonal elements for the standard errors works flawlessly regardless of the model's dimensions.

```r
summary.mylm(model2)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education + age, data = SLID)
##
## Coefficients:
##               Estimate Std. Error z value  Pr(>|z|)
## (Intercept) -6.0216529  0.6189238 -9.7292 < 2.2e-16 ***
## education    0.9014644  0.0357598 25.2089 < 2.2e-16 ***
## age          0.2570898  0.0089512 28.7212 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.249070
```

Interpretation of parameters: * **Intercept ($\hat{\beta}_0 \approx -6.02$):** The expected hourly wage for a theoretical person with 0 years of education and an age of 0. This holds no physical meaning but anchors the regression plane. * **Education ($\hat{\beta}_1 \approx 0.90$):** Holding age constant, each additional year of education increases the expected hourly wage by 0.90 units. * **Age ($\hat{\beta}_2 \approx 0.26$):** Holding education constant, each additional year of age increases the expected hourly wage by 0.26 units. * **Significance:** The z-tests for all variables yield p-values strictly close to 0, indicating that both education and age are highly significant predictors of wages.

c) Simple vs. Multiple Regression Estimates

The estimated parameter for `education` differs between the simple regression (Part 2) and the multiple regression (Part 3) due to **omitted variable bias** (confounding).

In the simple linear regression, the `education` coefficient absorbs both its direct effect on wages and the indirect effect of the omitted variable (`age`), since `age` and `education` are correlated (as seen in Part 1, $r = -0.106$). In the multiple linear regression, the coefficient isolates the **partial effect** of `education` on wages while explicitly controlling for `age`.

We can show these differences by running the individual models with `mylm` and comparing their coefficients:

```r
# Fit the individual simple regressions
mod_edu <- mylm(wages ~ education, data = SLID)
mod_age <- mylm(wages ~ age, data = SLID)

# Display the isolated vs partial effects
cat("Simple regression education effect:   ", mod_edu$coefficients["education"], "\n")
```

```
## Simple regression education effect:    0.7923091
```

```r
cat("Simple regression age effect:         ", mod_age$coefficients["age"], "\n\n")
```

```
## Simple regression age effect:          0.2331079
```

```r
cat("Multiple regression education effect: ", model2$coefficients["education"], "\n")
```

```
## Multiple regression education effect:  0.9014644
```

```r
cat("Multiple regression age effect:       ", model2$coefficients["age"], "\n")
```

```
## Multiple regression age effect:        0.2570898
```

“‘